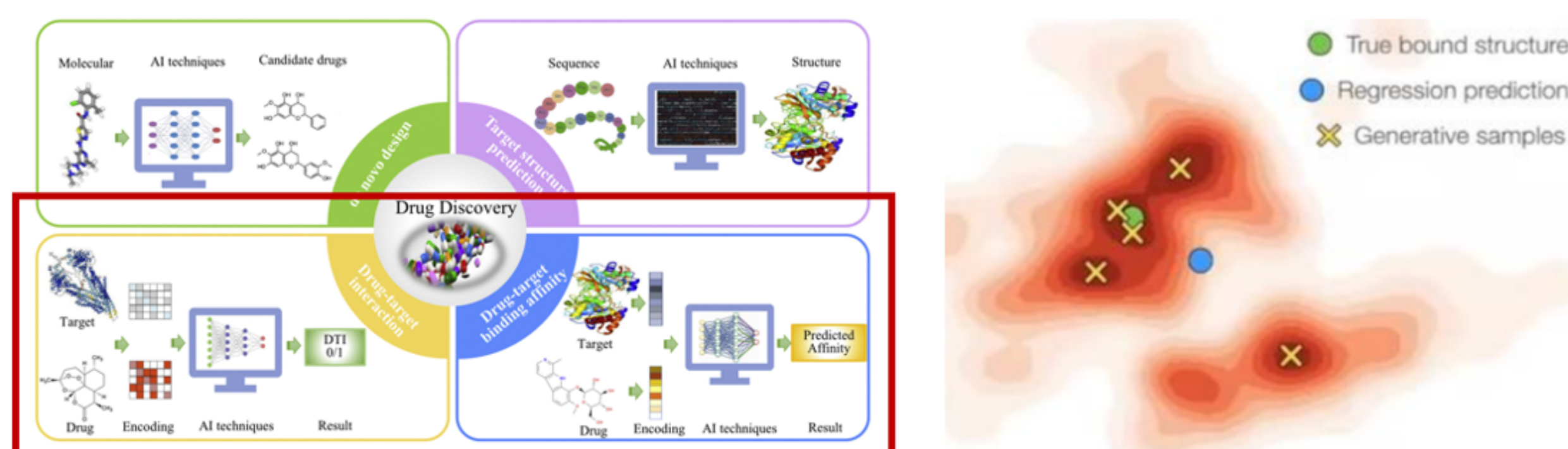
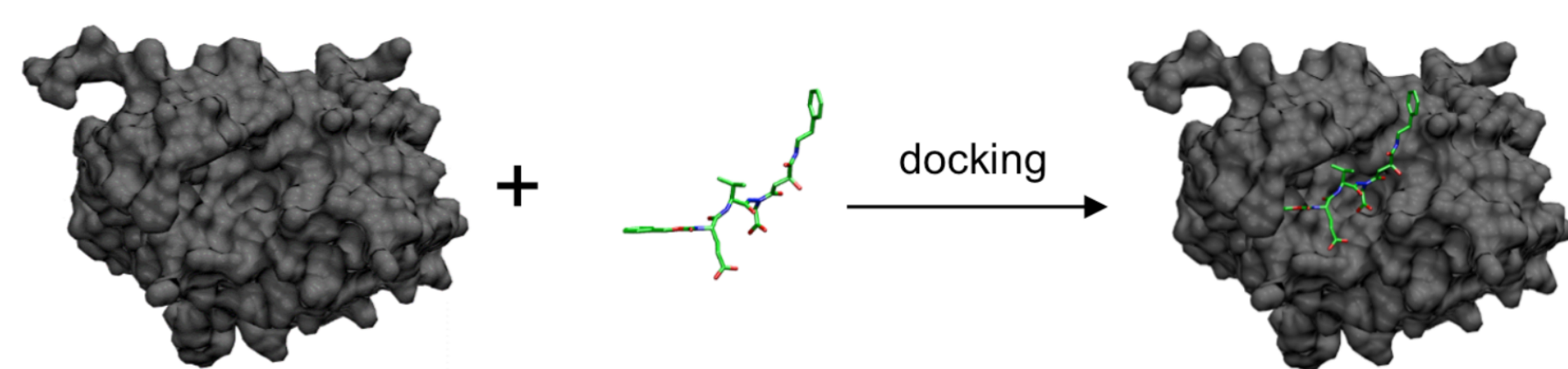


Abstract

- Artificial intelligence, through **generative models**, has emerged as a crucial tool in the realm of **drug discovery**.
- Diffusion models, which utilize **stochastic diffusion equations**, have proven successful in generating new data points to expand the molecular landscape for drug discovery.

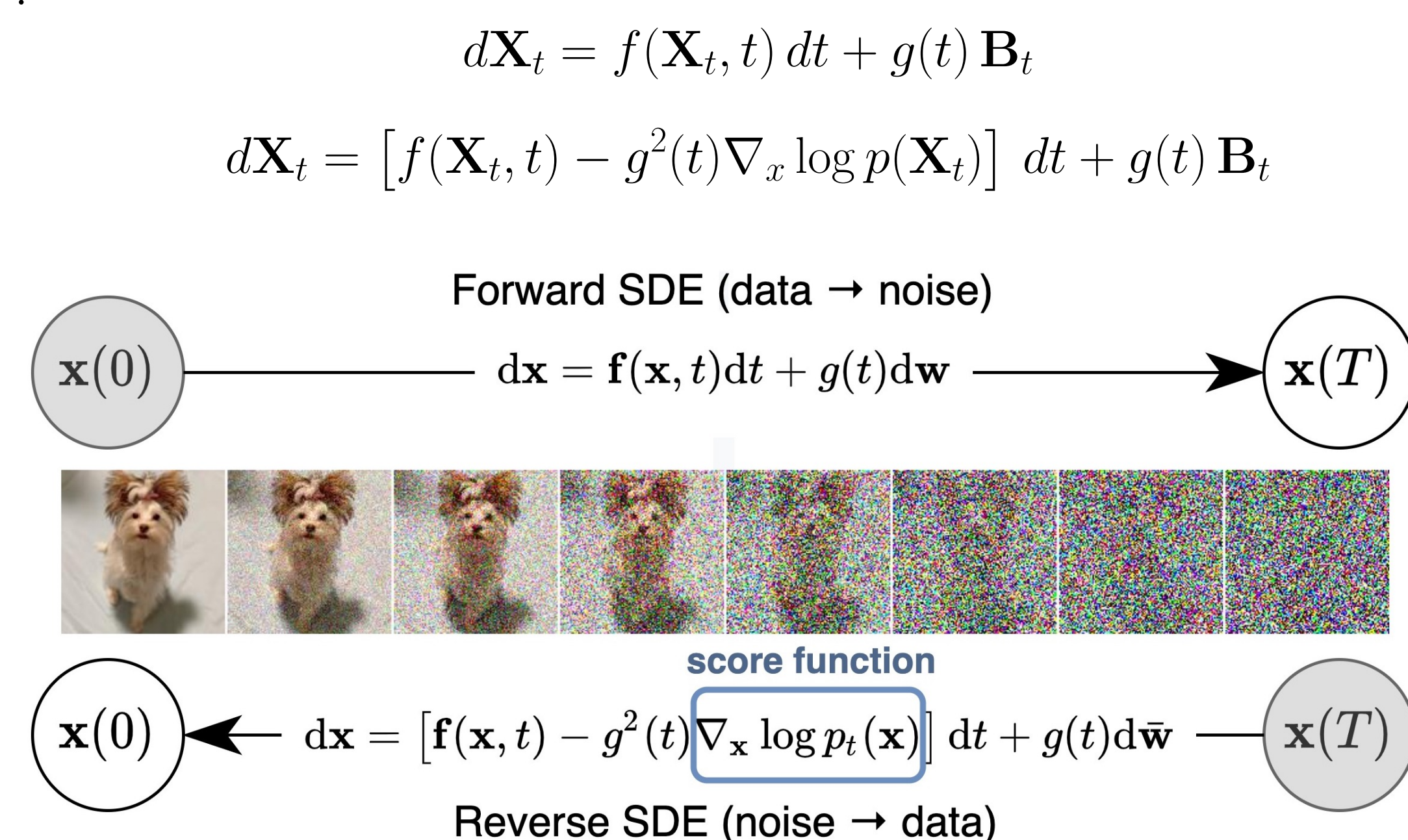


- The **DiffDock software** represents a significant innovation in **molecular docking**, a computational technique vital for predicting the binding interactions between small molecules (ligands) and target proteins and the ligand's pose including its position and orientation.
- Molecular docking is essential for determining the potential affinity and stability of the interaction between a ligand and its protein, facilitating the identification of compounds that can effectively bind to **protein sites linked to diseases**. It is crucial for the drug discovery process.

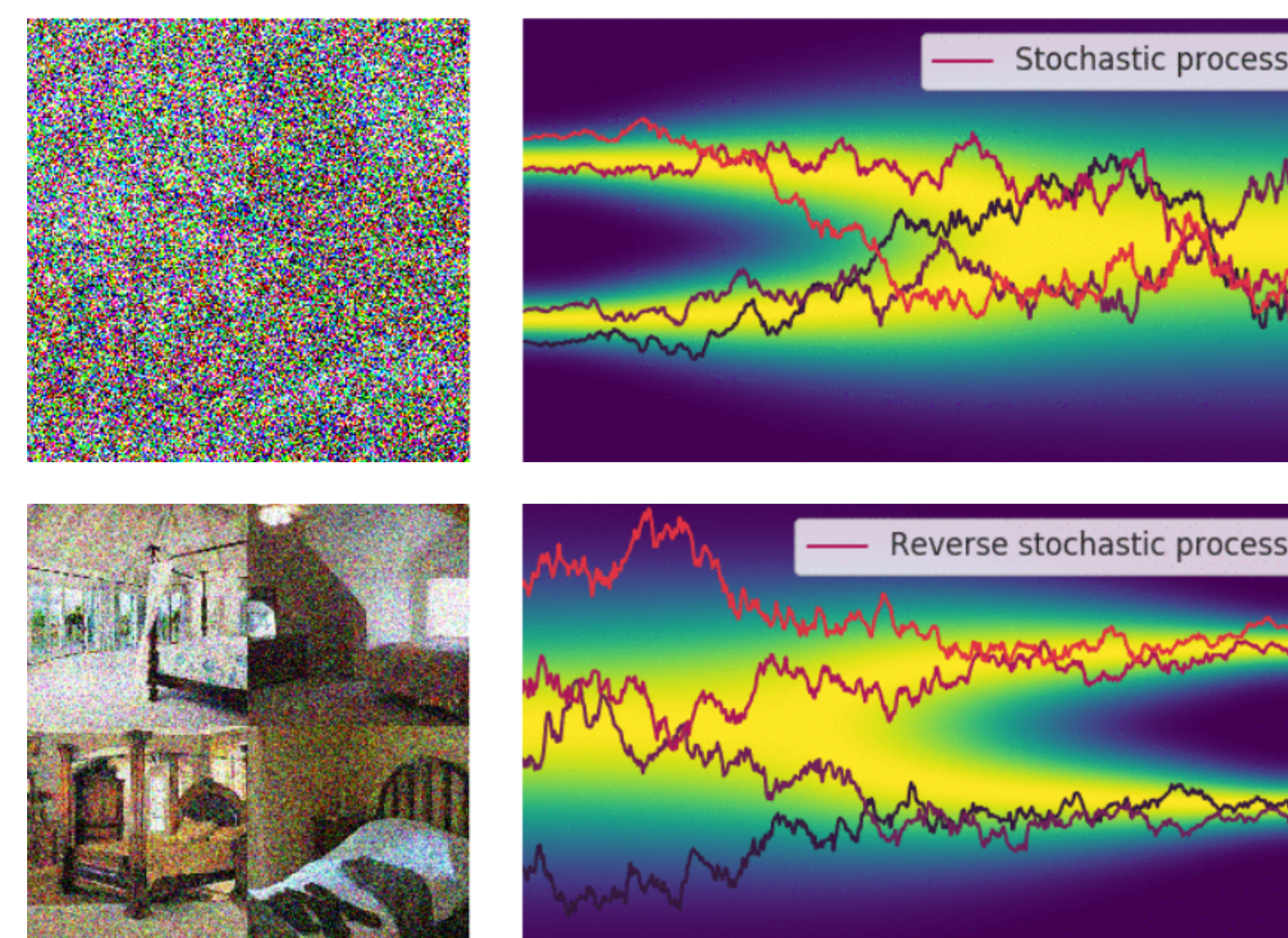


Focus on diffusion models with score-based models

- Diffusion models** work by gradually transforming an initial data distribution (noise) into a target data distribution, through an iterative process. This process is guided by models that **learn to reverse these diffusions**, allowing the **generation of samples from noise** by following the **inverse steps of diffusion**.
- Score models** learn score functions (gradients of logarithmic probability densities) over many data distributions perturbed by noise through **Stochastic Differential Equation (SDE)**, and then generate samples via the reverse SDE.
- SDEs and their reverse counterparts (reverse SDEs) commonly take the following forms :



Process of Diffusion and Reverse Diffusion in Score-based Generative Models



Stochastic Trajectory of Diffusion Models :
from Random Noise to Structured Representation with Score Functions

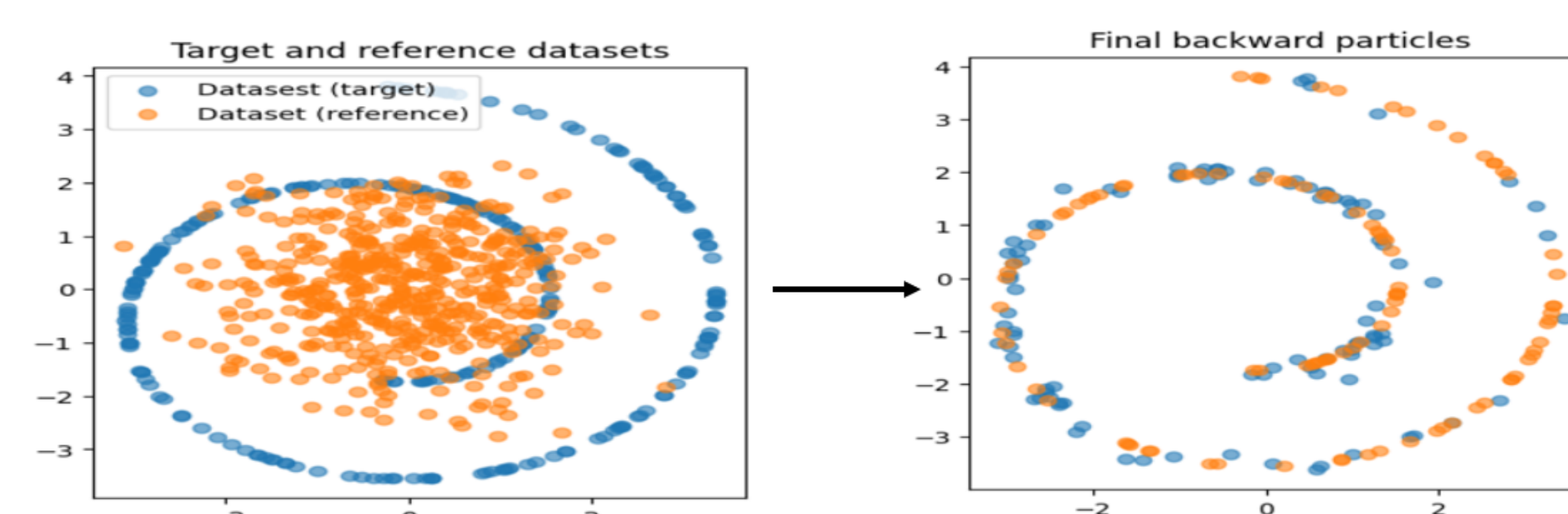
We restrict ourselves to a simple two-dimensional setting and want to generate a spiral distribution from a gaussian distribution

- Forward Noising Process with the following SDE :

$$d\mathbf{X}_t = -\frac{1}{2}\mathbf{X}_t dt + d\mathbf{B}_t, \quad \mathbf{X}_0 \sim \pi,$$

- Backward Denoising Process using the reverse SDE :

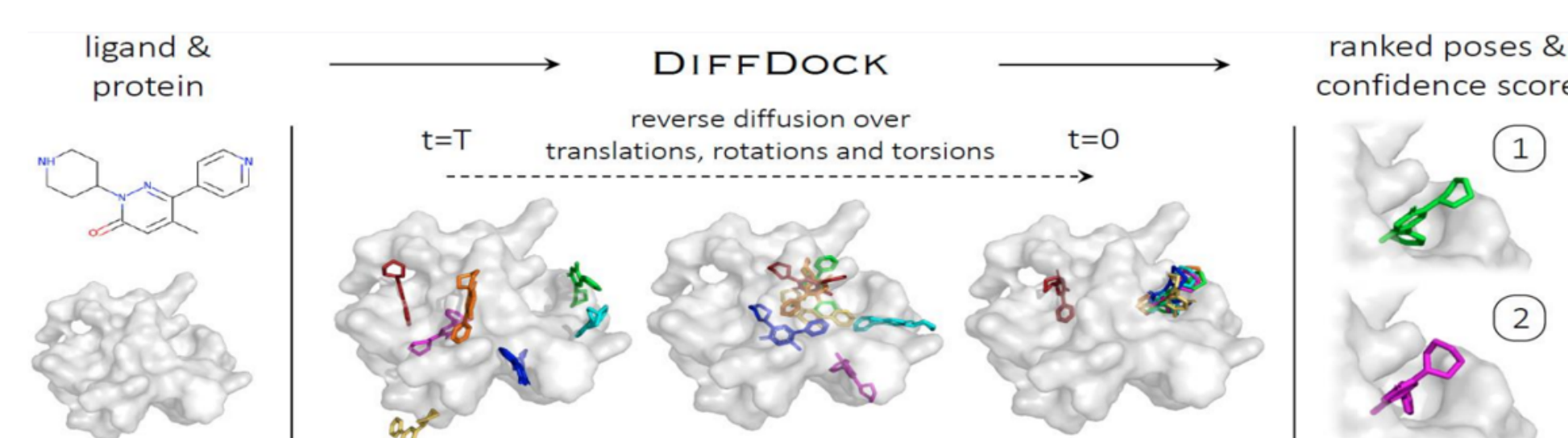
$$d\mathbf{X}_t = \left(\frac{1}{2}\mathbf{X}_t + \nabla \log p_{T-t}(\mathbf{X}_t) \right) dt + d\mathbf{B}_t, \quad \mathbf{X}_0 \sim \mathcal{N},$$



Reverse Diffusion Process

Diffdock and results

- DiffDock uses diffusion models, to simulate the binding process between ligands and proteins. By temporally reversing stochastic diffusion equations, DiffDock generates **predictions about the optimal position and orientation** of ligand molecules relative to the target protein.



We use the following SDE to generate independently position, orientation and torsion of the ligand on the protein :

$$d\mathbf{X}_t = \sqrt{\frac{d\sigma^2(t)}{dt}} d\mathbf{B}_t \quad \text{where} \quad \sigma^2 = \sigma_{tr}^2, \sigma_{rot}^2, \text{ or } \sigma_{tor}^2 \quad \text{for } T(3), SO(3), \text{ and } SO(2)^m$$

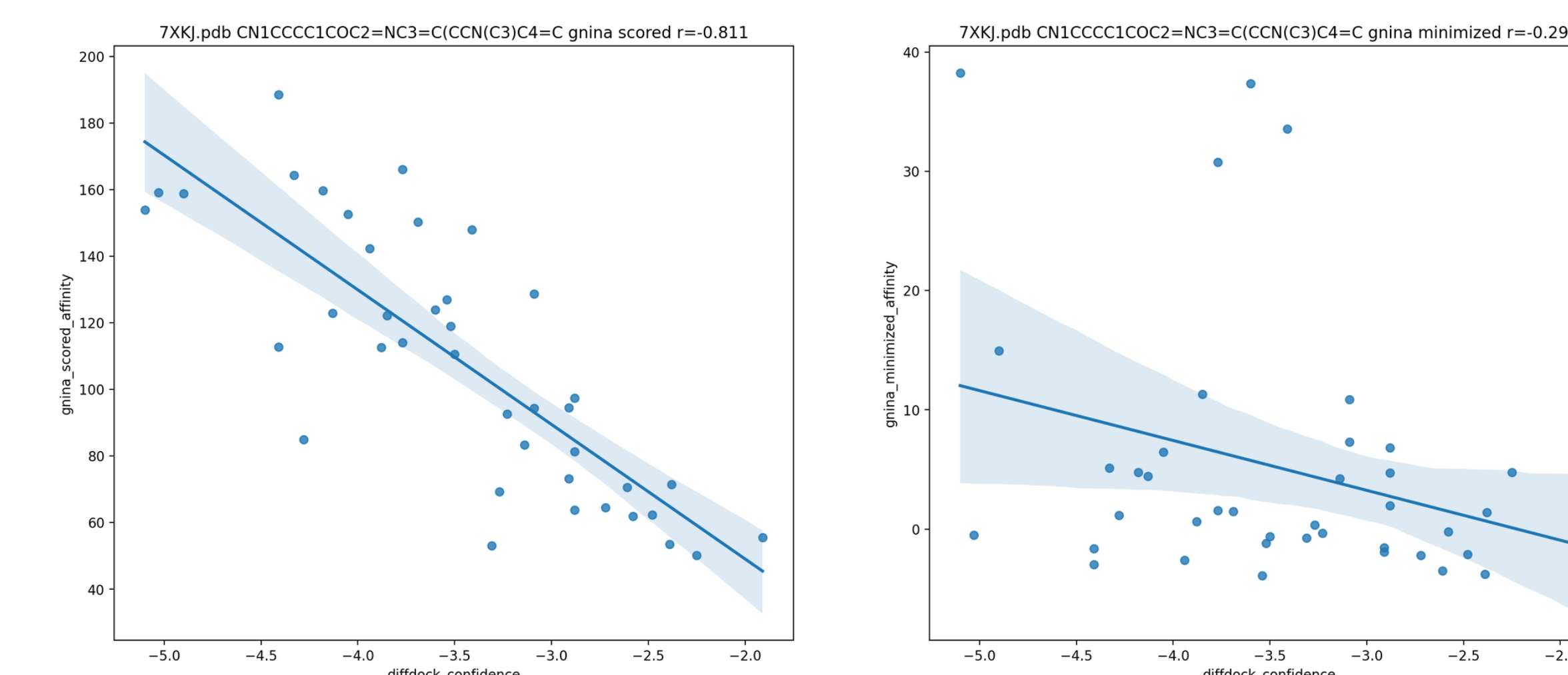


For our DiffDock test, we will focus on **KRAS** and **Adagrasib** molecules

- KRAS** is a gene encoding the K-Ras protein, crucial in regulating cell growth and differentiation. Mutations in KRAS are prevalent in cancer.
- Adagrasib** is a small molecule inhibitor targeting the mutated form of K-Ras. Despite challenges due to the absence of a clear binding pocket, Adagrasib has shown promising anti-tumor activity in clinical studies.

Diffdock confidence score

- DiffDock is a two-step process : we train the diffusion model and a **confidence score** model based on the root mean square distance (RMSD) between the structures.

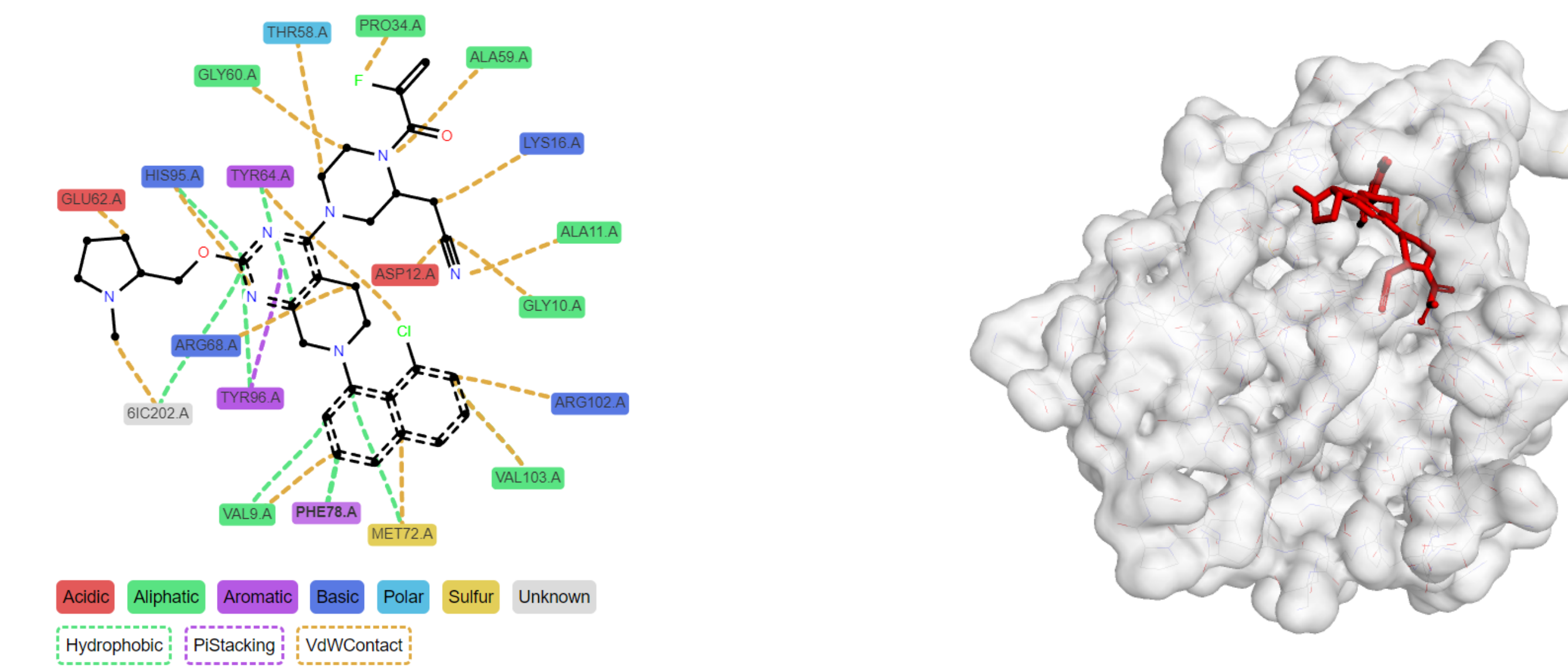


Comparison of DiffDock Confidence with Gnina Predicted Affinity

- We compare our confidence scores with others tools like GNINA affinity scores.
- Then we **optimize the ligand's pose** by locally **minimizing the potential energy of the protein-ligand binding** using the GNINA software.
- GNINA employs evolutionary algorithms to explore various ligand conformations and generate a population of ligand poses, evaluate them based on criteria such as energy and interactions. Then, it optimizes the ligand-receptor system energy using numerical optimization methods like gradient descent to find the most stable and favorable pose.

Our DiffDock results

- The different types of **chemical interactions** could be visualized.
- Such information is crucial for predicting the ligand's **affinity** and **specificity**, as well as for guiding the design of chemical modifications aimed at **enhancing interaction with the target protein**.



Best confidence score result achieved with DiffDock using KRAS and Mirati

References

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